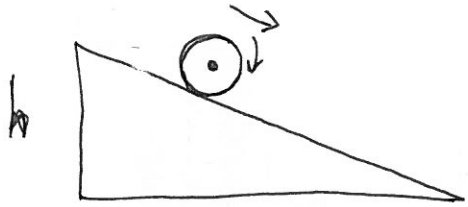
P1



Assume good friction $\Rightarrow$ rolling without slipping.

Energy is conserved.

$$E_i = Mgh$$

$$E_f = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I \omega^2$$

but $\omega = \frac{v_{cm}}{R} \Rightarrow E_f = \frac{1}{2}\left(M + \frac{I}{R^2}\right) v_{cm}^2$

$$Mgh = \frac{1}{2}\left(M + \frac{I}{R^2}\right) v_{cm}^2$$

$$v_{cm} = \sqrt{\frac{2Mgh}{M + \frac{I}{R^2}}}$$

Solid cylinder : $I = \frac{1}{2} M R^2 \Rightarrow v_{cm} = \sqrt{\frac{2Mgh}{M + \frac{1}{2}M}} = \sqrt{\frac{4}{3} gh}$

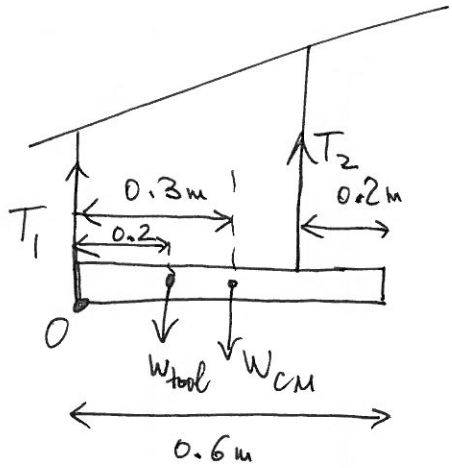
Cylindrical Shell : $I = MR^2 \Rightarrow v_{cm} = \sqrt{\frac{2Mgh}{M+M}} = \sqrt{gh}$

$$v_{cm}^{cylinder} > v_{cm}^{shell}$$

If there is <u>no friction</u>, then cylinders do not rotate

$\Rightarrow Mgh = \frac{1}{2} M v_{cm}^2 \Rightarrow v_{cm} = \sqrt{2gh}$ for both of them.

P2



$W_{CM} = 50.0\ N$

$W_{tool} = 25.0\ N$

Equilibrium: no forces along x axis.

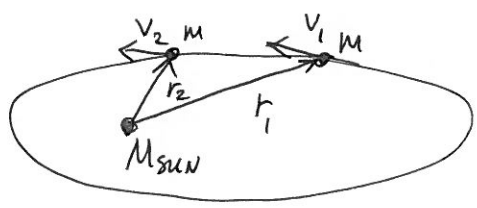
along y: $T_1 + T_2 - W_{tool} - W_{CM} = 0$

torque about O: $-0.2\, W_{tool} - 0.3\, W_{CM} + 0.4\, T_2 = 0$

$$\Rightarrow T_2 = \frac{0.2(25.0) + 0.3\cdot(50.0)}{0.4} = 50.0N$$

$$T_1 = (25.0) + (50.0) - T_2 = 25.0N$$

P3



$G = 6.67\cdot 10^{-11}\ \frac{N\cdot m^2}{kg^2}$

$M_{SUN} = 1.99\cdot 10^{30}\ kg$

Use conservation of energy:

$$E_1 = \frac{1}{2} m V_1^2 - \frac{G M_{SUN}\, m}{r_1}$$

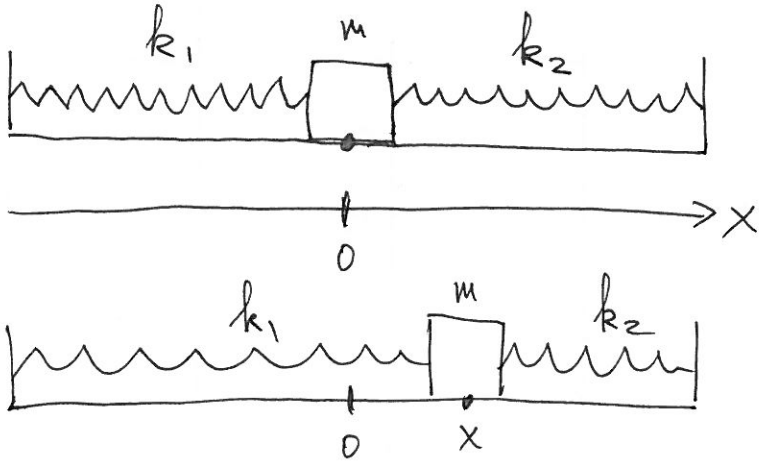
$$E_2 = \frac{1}{2} m V_2^2 - \frac{G M_{SUN}\, m}{r_2}$$

$$E_1 = E_2 \Rightarrow \frac{1}{2} m V_1^2 - \frac{GMm}{r_1} = \frac{1}{2} m V_2^2 - \frac{GMm}{r_2}$$

Divide by m: $V_2^2 = V_1^2 - \frac{2GM}{r_1} + \frac{2GM}{r_2} = V_1^2 - 2GM\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

$V_2 = 6.8\cdot 10^4\ m/s$

P4



(a) $E_i = \frac{1}{2} k_1 x^2 + \frac{1}{2} k_2 x^2$ as the two springs are deformed by the same X.

Block will have maximum speed when it passes through $x=0$ as then the springs are <u>not</u> deformed. At that instant

$$E_f = \frac{1}{2} m V^2 \Rightarrow \frac{1}{2} m v^2 = \frac{1}{2}(k_1+k_2) x^2$$

$$V = \sqrt{\frac{k_1+k_2}{m}}\, X = 0.230 \text{ m/s}$$

(b)

Restoring forces from the two springs act in the <u>same direction</u>

$$F_1 = -k_1 x$$
$$F_2 = -k_2 x$$

$$\Rightarrow \quad ma = -k_1 x - k_2 x = -(k_1+k_2) x$$

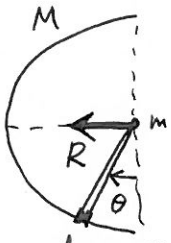
$$a = \frac{d^2x}{dt^2} = -\frac{k_1+k_2}{m} x = -\omega^2 x$$

$$\Rightarrow \omega = \sqrt{\frac{k_1+k_2}{m}}, \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k_1+k_2}}$$

$\omega = 4.61 \text{ s}^{-1}$ $\qquad$ $T = 1.36 \text{ s}$

P5

(a)



semi-circle $\Rightarrow L = \pi R$, $R = \frac{L}{\pi}$

$$dM = \frac{R d\theta}{L} M = \frac{d\theta}{\pi} M$$

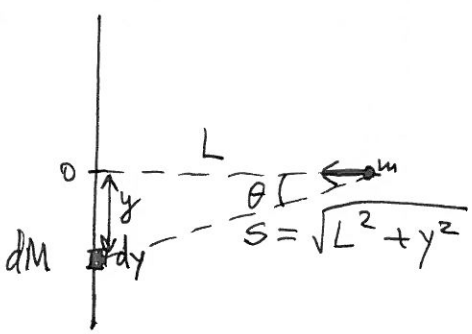
$$dF = \frac{G m\, dM}{R^2} \sin\theta = \frac{G m M}{\pi R^2} \sin\theta\, d\theta$$

we only need horizontal component as y-component is zero by symmetry

$$F = \int dF = \frac{G m M}{\pi R^2} \underbrace{\int_0^{\pi} \sin\theta\, d\theta}_{\int_0^{\pi} d(-\cos\theta) = (-1)\left[\cos\pi - \cos 0\right] = 2.} = \frac{2}{\pi} \frac{G m M}{R^2}$$

$$R = \frac{L}{\pi} \Rightarrow F = 2\pi \frac{G m M}{L^2}.$$

(b)



$$dM = \frac{M}{L} dy$$

$$dF = \frac{G m\, dM}{s^2} \cos\theta$$ (only horizontal component is finite!)

$$\cos\theta = \frac{L}{s} = \frac{L}{\sqrt{L^2+y^2}}$$

$$F = \int \frac{G m M}{L s^2} \cos\theta\, dy = \frac{G m M}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{L}{(L^2+y^2)^{3/2}} dy =$$

$y = Lu$

$$= \frac{G m M}{L^2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{du}{(1+u^2)^{3/2}} = \frac{G m M}{L^2} \left. \frac{2u}{\sqrt{1+u^2}} \right|_{-1/2}^{1/2} = \sqrt{\frac{4}{5}} \frac{G m M}{L^2}.$$